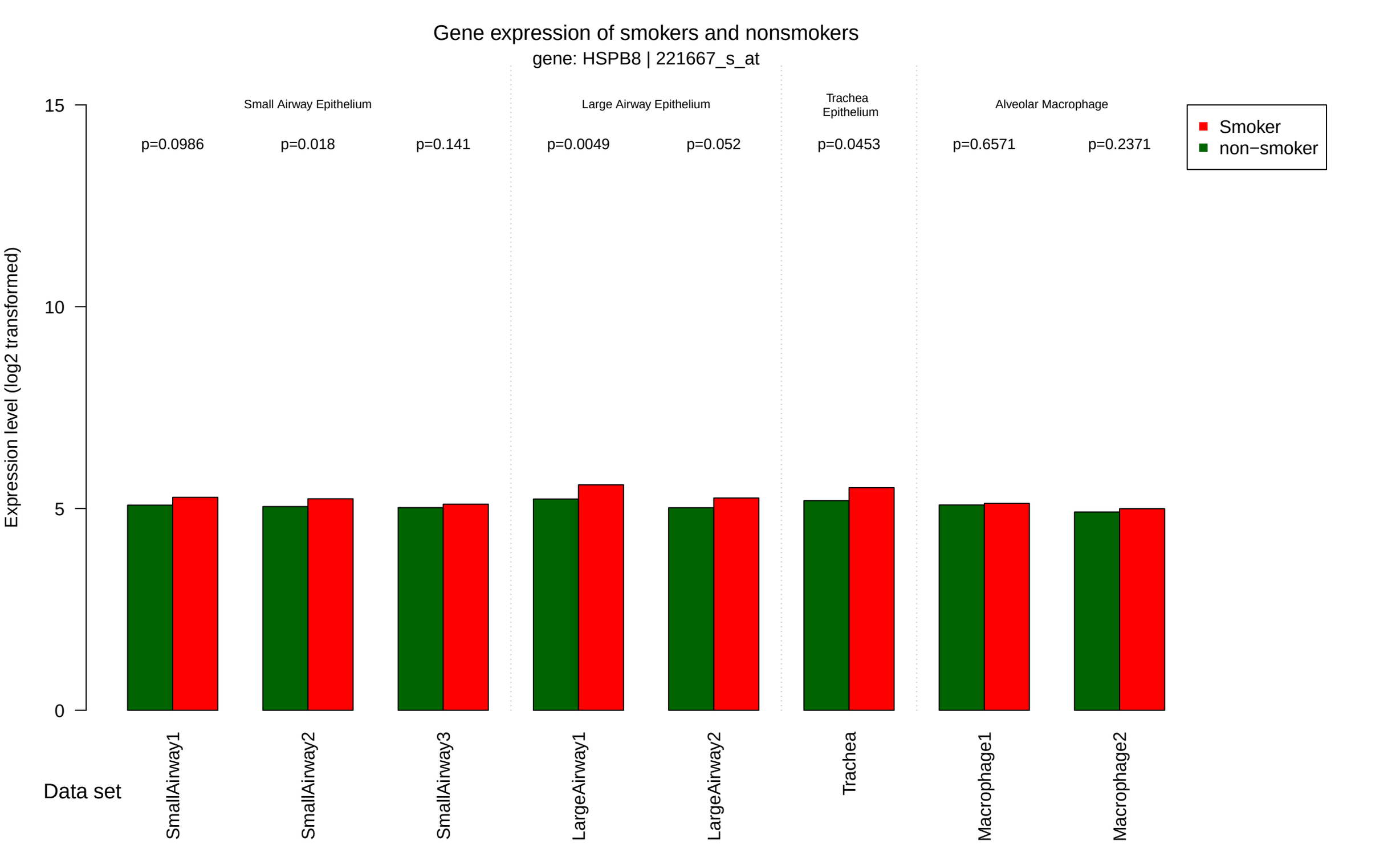
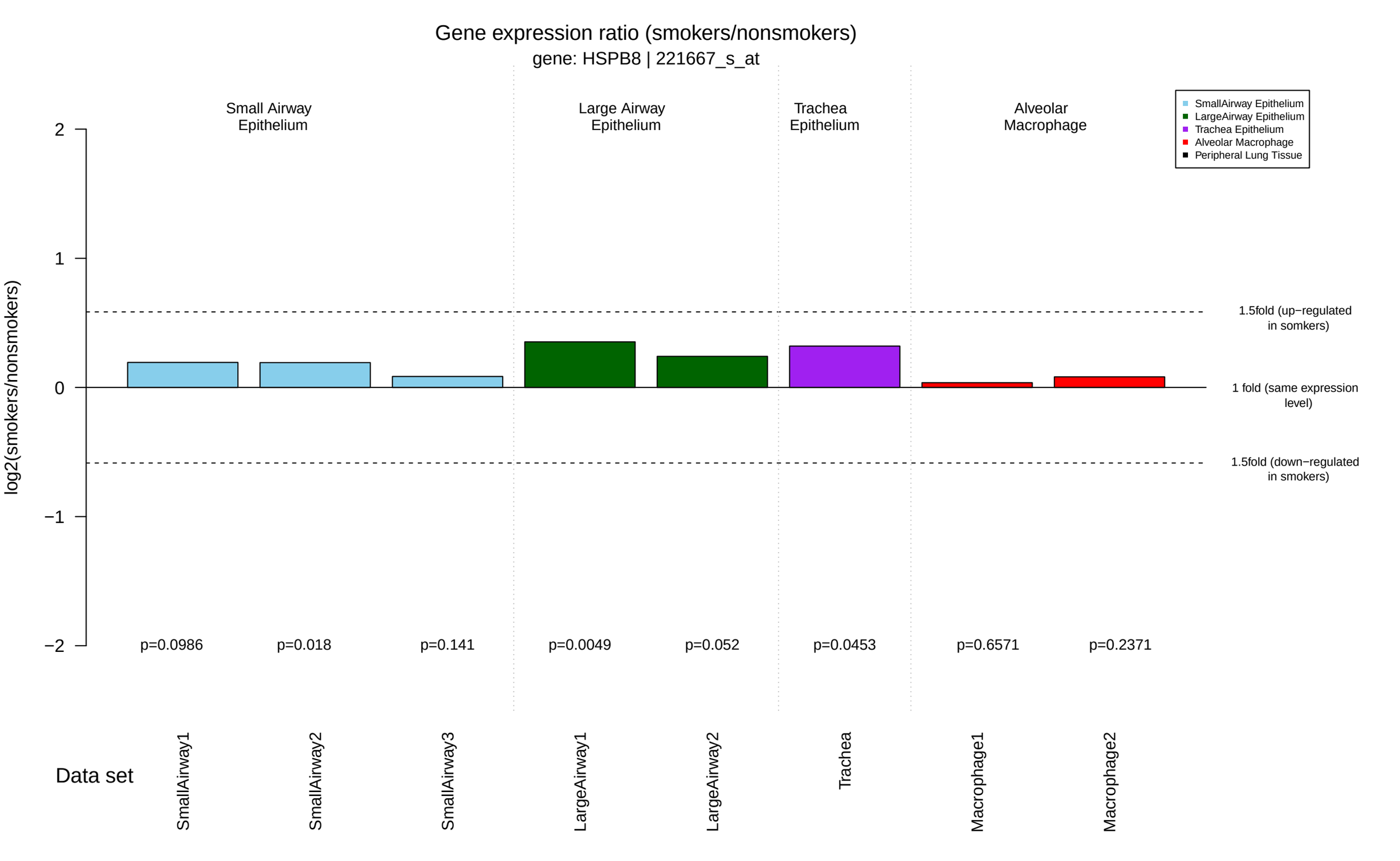
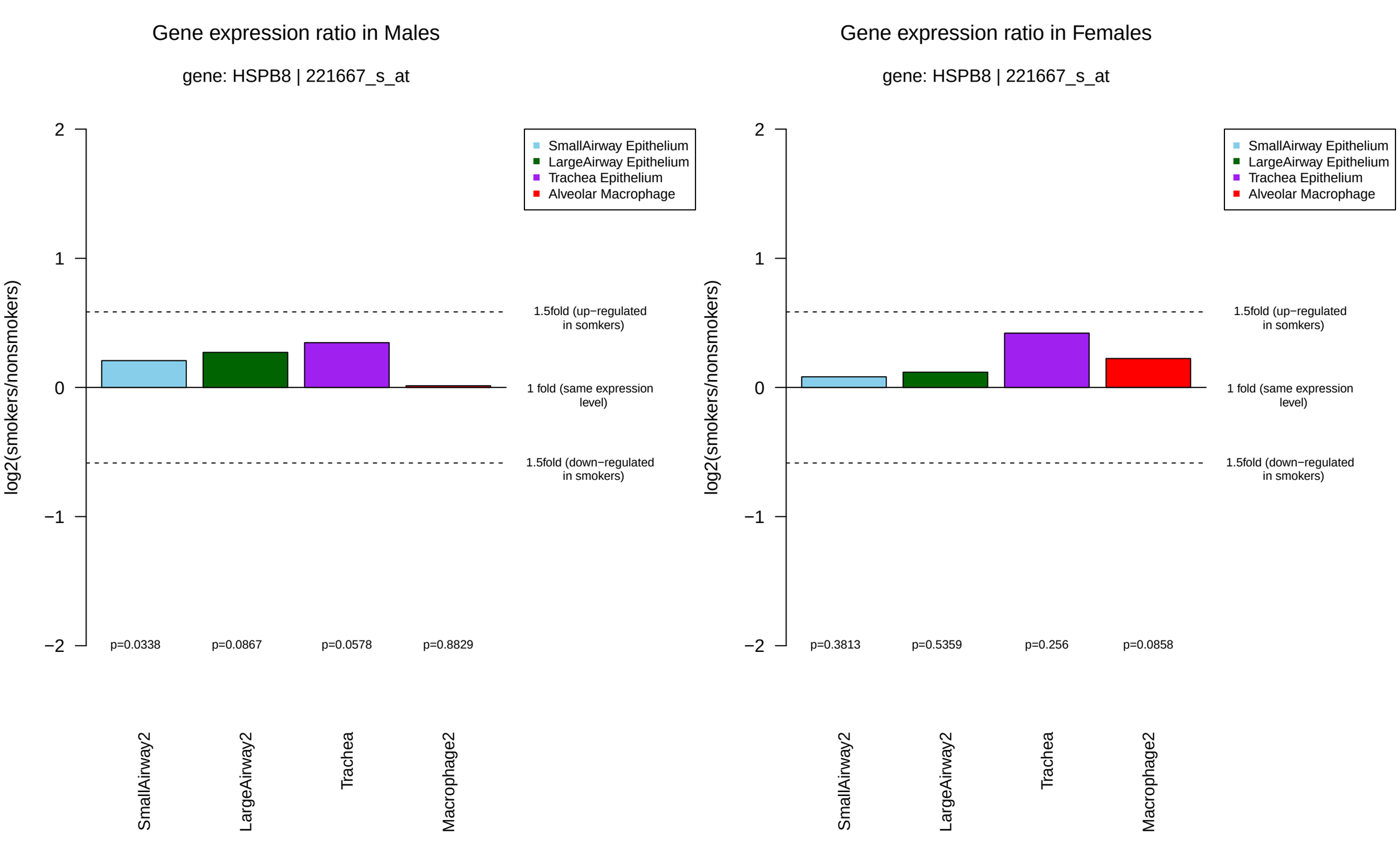
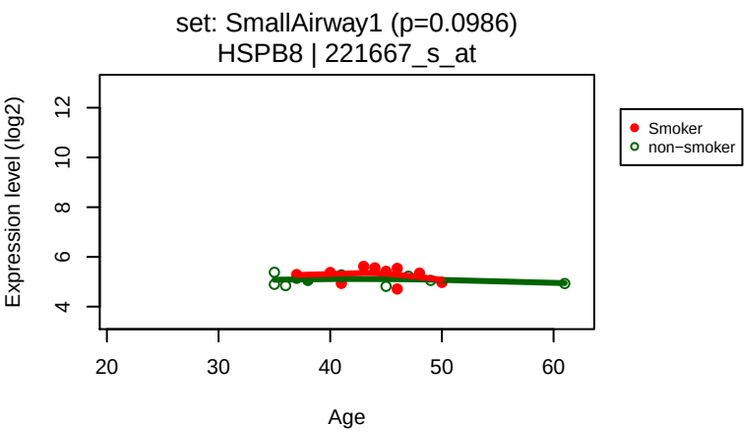
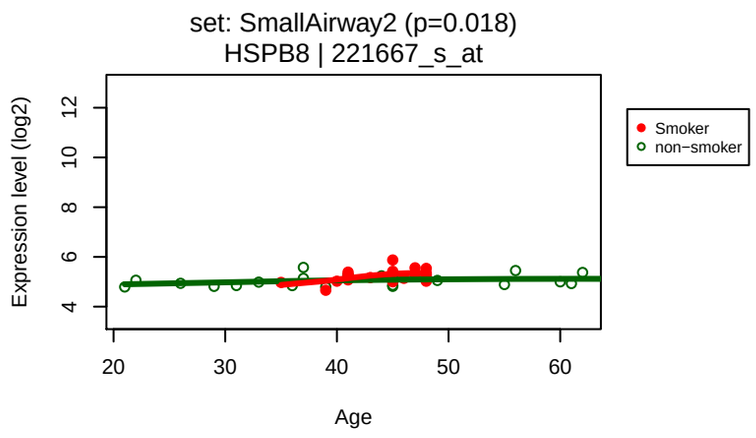
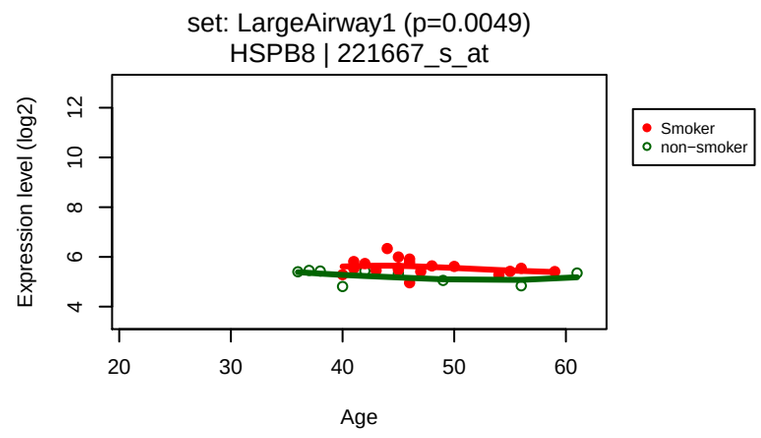

Gene expression of smokers and nonsmokers
gene: HSPB8 | 221667_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0986
p=0.018
p=0.141
p=0.0049
p=0.052
p=0.0453
p=0.6571
p=0.2371
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

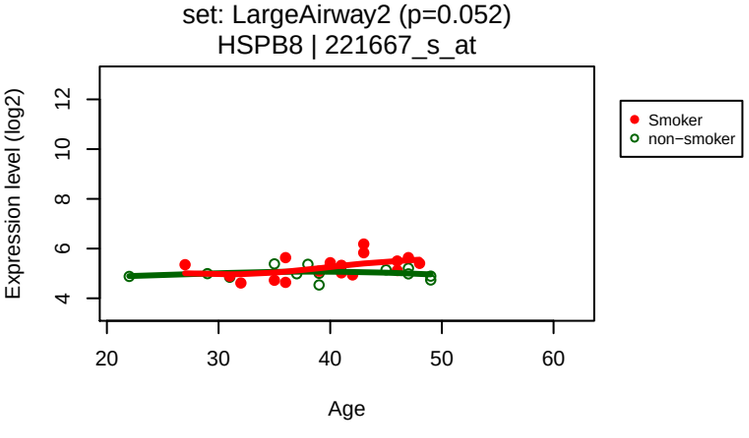

Gene expression ratio (smokers/nonsmokers)
gene: HSPB8 | 221667_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0986
p=0.018
p=0.141
p=0.0049
p=0.052
p=0.0453
p=0.6571
p=0.2371
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

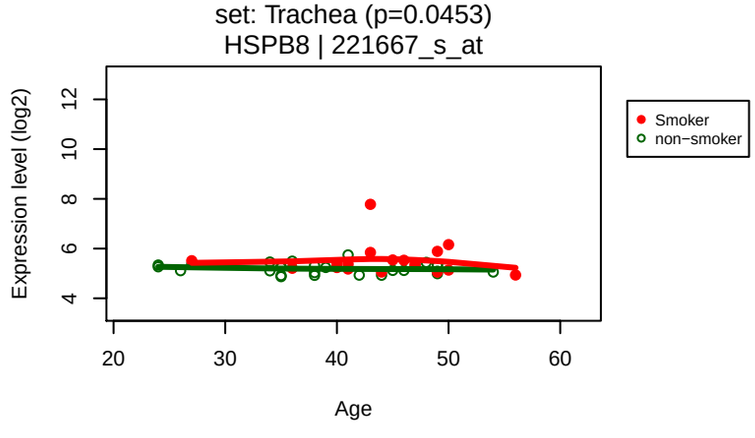

Gene expression ratio in Males
gene: HSPB8 | 221667_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0338
p=0.0867
p=0.0578
p=0.8829
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HSPB8 | 221667_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3813
p=0.5359
p=0.256
p=0.0858
SmallAirway2
LargeAirway2
Trachea
Macrophage2

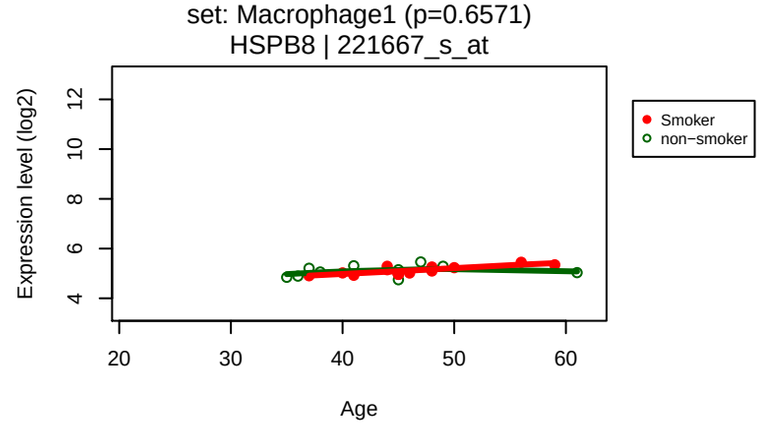

set: SmallAirway1 (p=0.0986)
HSPB8 | 221667_s_at
Expression level (log2)
Age
Smoker
non−smoker

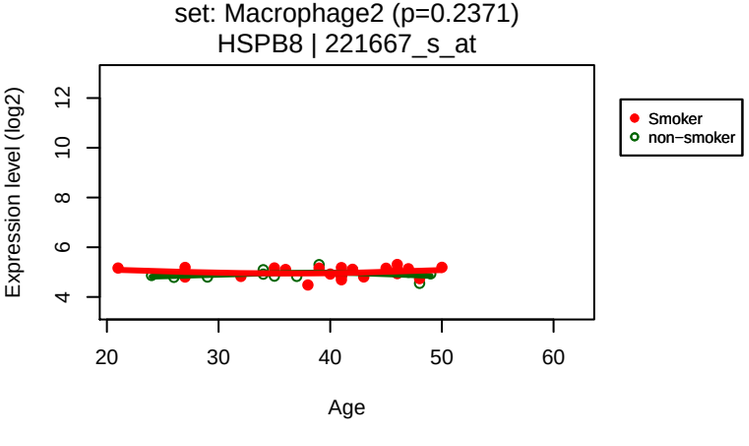

set: SmallAirway2 (p=0.018)
HSPB8 | 221667_s_at
Expression level (log2)
Age
Smoker
non−smoker

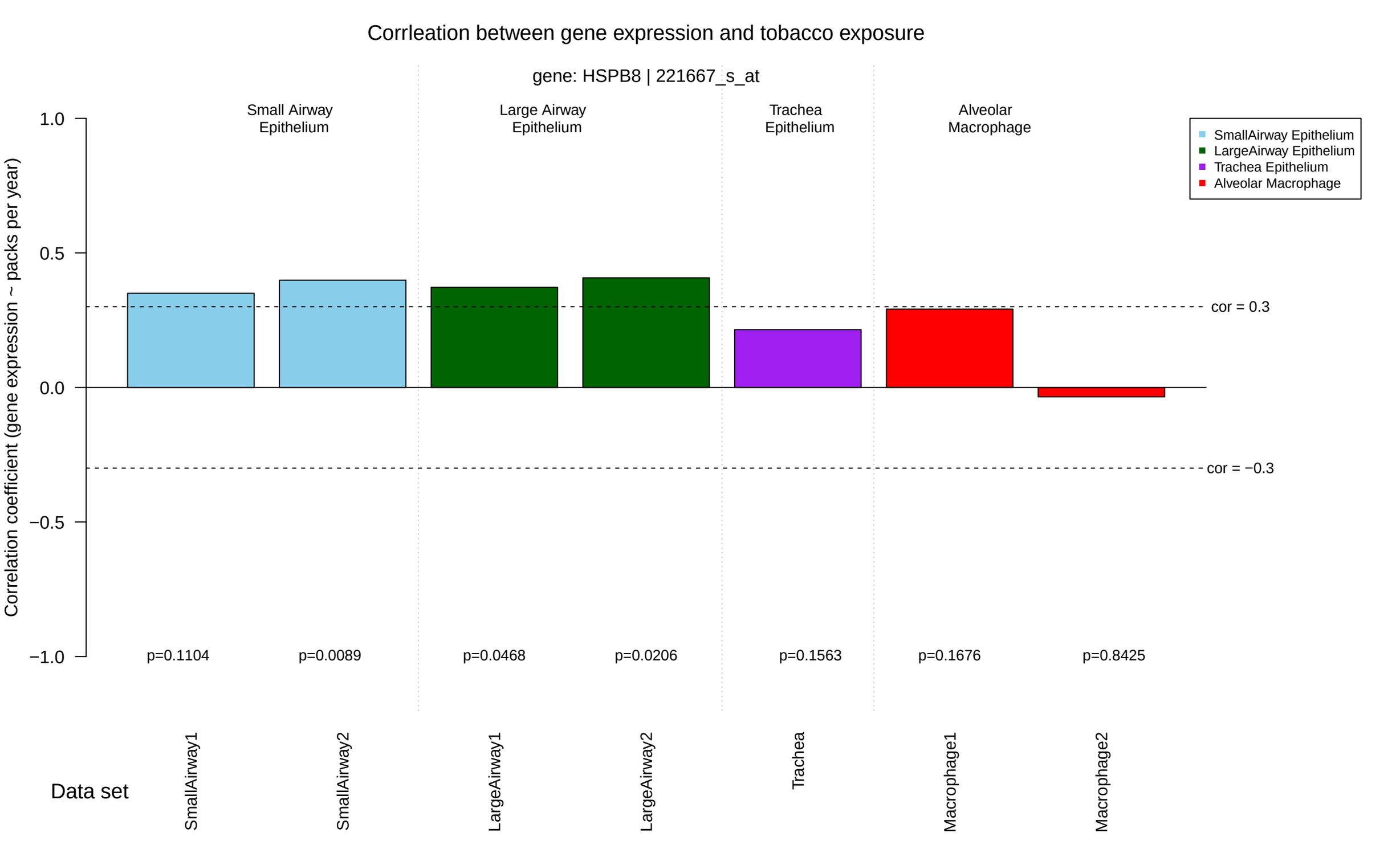

set: LargeAirway1 (p=0.0049)
HSPB8 | 221667_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.052)
HSPB8 | 221667_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0453)
HSPB8 | 221667_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6571)
HSPB8 | 221667_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2371)
HSPB8 | 221667_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HSPB8 | 221667_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1104
p=0.0089
p=0.0468
p=0.0206
p=0.1563
p=0.1676
p=0.8425
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2